TRACING THE TECHNICAL EVOLUTION OF ADVERSARIAL EXPLOITS AND ROBUSTNESS

The Lineage of Machine Learning Attacks

ARJUN T.U | 2026-06-22

Adversarial Attacks Have Shifted from White-Box to Black-Box Scenarios

Advancement towards realistic, restrictive threat models over the past decade

Phase 1

White-Box Dominance

Fast Gradient Sign Method (2014)

Projected Gradient Descent (2017)

Phase 2

Soft-Label & Transferability

Transfer Attack (2016)

Phase 3

Hard-Label Black-Box

Boundary Attack (2018, 2019)

Optmization Attack (2019, 2021)

HopSkipJump Attack (2020)

Triangle Attack (2022)

Small Query
Black-box Adversarial Attack (2024)

EXPERIMENTAL SETUP & DEMO SPECIFICATIONS

CIFAR-10 Binary Classification: VictimNet Architecture

Dataset: CIFAR-10 (3072-dim)

Scope: 2-class slice focusing on Airplane vs. Automobile differentiation.

VICTIMNET ARCHITECTURE (CNN)

- **Layer 1:** Conv (32 filters, 3×3) + ReLU + 2×2 MaxPool (32×32 16×16)
- **Layer 2:** Conv (64 filters, 3×3) + ReLU + 2×2 MaxPool (16×16 8×8)
- **Layer 3:** Conv (64 filters, 3×3) + ReLU + 2×2 MaxPool (8×8 4×4)
- **Output:** Fully Connected (Linear) layer. 1024-dim flattened 2 Logits.

VISUALIZING THE TARGET CLASSES

INPUT: 32x32 RGB Image (3,072 features)

TARGET: Binary Logic [Airplane, Automobile]

Understanding Gradients & Cross-Entropy Optimization

The Gradient (∇_θL) is a vector of 58,370 numbers in this architecture, representing the slope for every specific weight.

CORE FUNCTION:

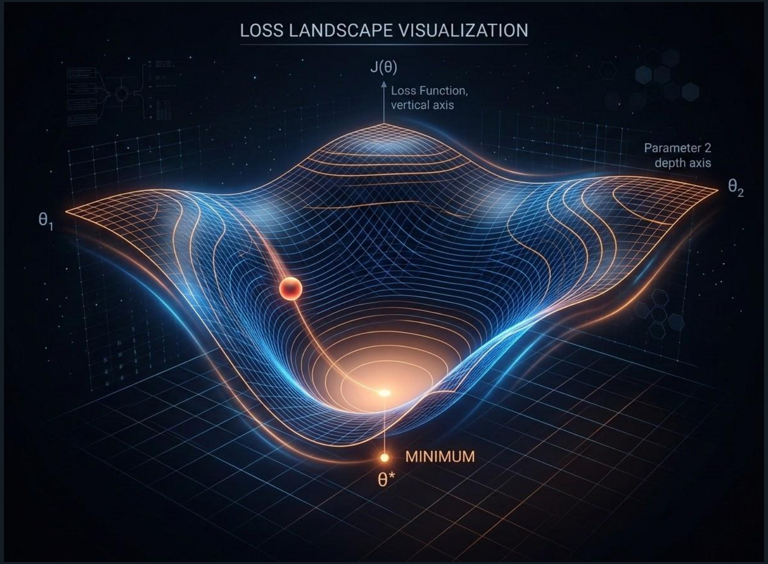
It answers: "If I tweak this specific filter weight, will the model's error go up or down?"

STOCHASTIC GRADIENT DESCENT:

$$\theta\_new = \theta\_old - \eta \cdot \nabla\_\theta L$$

The weights (θ) are updated by moving in the opposite direction of the gradient to minimize the loss.

VISUALIZING THE LOSS LANDSCAPE



MATHEMATICAL FOUNDATIONS OF ATTACK

Weaponizing the Gradient: From Training to Attack

Dual Nature of Gradients: In training, we update weights. In an attack, we keep weights (θ) **constant** and update the image (x) instead.

ATTACK CORE FUNCTION:

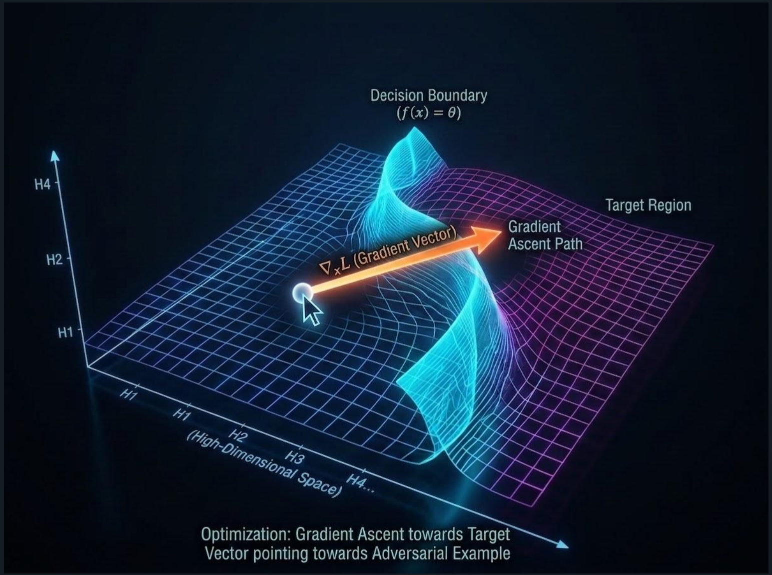
Instead of asking how to minimize loss, the attacker asks: "How can I slightly change the input pixels to **maximize** the error?"

ADVERSARIAL GRADIENT ASCENT (FGSM):

$$x_{adv} = x + \varepsilon \cdot \text{sign}(\nabla_x L)$$

By following the gradient of the loss with respect to the input ($\nabla_x L$), the attacker finds the most "vulnerable" direction to push the image across the decision boundary.

VISUALIZING THE ATTACK VECTOR (INPUT SPACE)



PHASE 2: CROSS-MODEL VULNERABILITY

Transfer Attack: The Black-Box Sweep

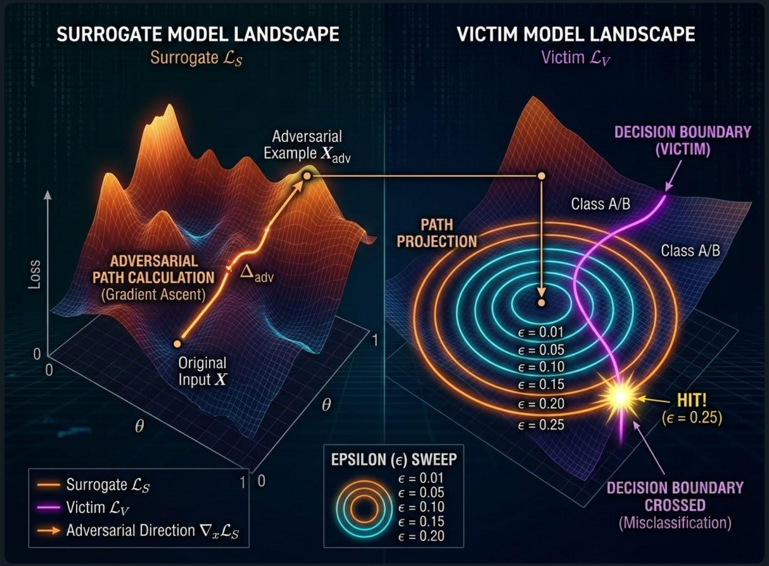
Concept: The Transfer Attack uses PGD as its engine but applies it to a black-box setting where gradients are hidden.

THE ATTACK LOGIC:

- Runs PGD on a **surrogate model** (white-box) to generate adversarial directions.
- **Budget Sweep:** Since the victim's ε is unknown, the attacker sweeps ε (e.g., 0.01 to 0.25).
- Stops at the **minimum ε** that successfully flips the label on the victim model.

This exploits the fact that adversarial directions often **transfer** across different architectures trained on the same data.

VISUALIZING THE EPSILON SWEEP & TRANSFERABILITY



PHASE 3: HARD-LABEL CONSTRAINTS

Finding the Initial Seed: Binary Search

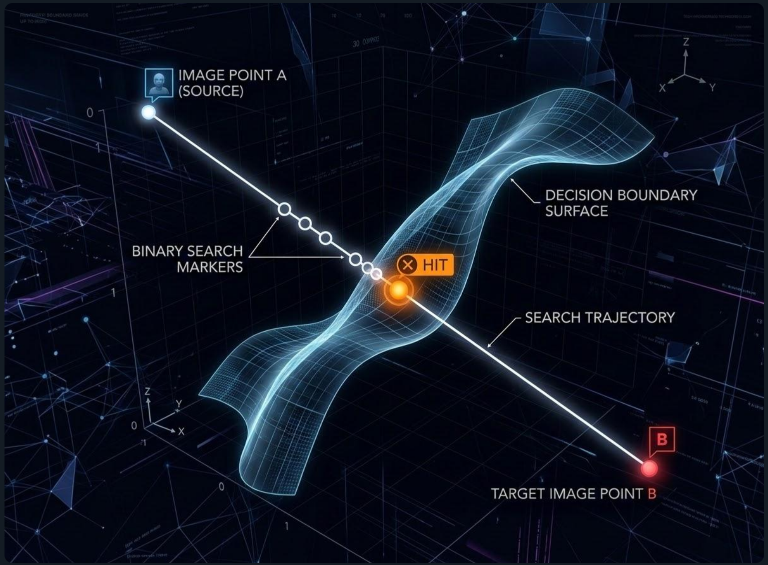
Optimization-Based Attacks: Most advanced black-box attacks require a starting "seed" adversarial point.

THE SEARCH ALGORITHM:

- Select a Target Image from a different class to act as the initial anchor.
- Binary Search: Iteratively explore the line between the clean image and the target image.
- Identify the nearest adversarial image closest to the boundary space

This process reduces a 3072-dimensional space (for a 32×32 RGB image) into a simple 1D line optimization problem.

VISUALIZING THE BOUNDARY INTERSECTION (BINARY SEARCH)



EVOLUTIONARY OVERVIEW

The Lineage of Machine Learning Attacks

TIER 1: WHITE-BOX
FGSM (2014) → PGD (2017)
Single-step vs iterative gradient steps. Foundations of vulnerability.

TIER 2: THE BLACK-BOX LEAP
Transfer Attack (2016)
Zero-query success using surrogate model boundaries.

TIER 3: LOCAL SEARCH
Boundary Attack (2018)
Pure random walk along the 'wall'. Guaranteed but slow.

TIER 4: PATH OPTIMIZATION
OPT → Sign-OPT (2020)
Directional optimization. Sign-only probes for efficiency.

TIER 5: NORMAL ESTIMATION
HopSkipJump (2020)
Analytic perpendicular vector via Monte Carlo probing.

TIER 6: SUBSPACE SPEED
Triangle Attack (2022)
Low-frequency DCT planes. Geometric Law of Sines (2 queries).

TIER 7: THE HYBRID FUSION (STATE OF THE ART)
Biased-Boundary (2019) → SQBA (2024)
The ultimate blend: Transfer speed + HSJ accuracy. Uses Beta Switch to adapt to surrogate quality dynamically.

EVOLUTIONARY TRAJECTORY: GRADIENT ACCESS → QUERY EFFICIENCY → HYBRID ADAPTATION

Boundary Attack: "Walking Along the Wall"

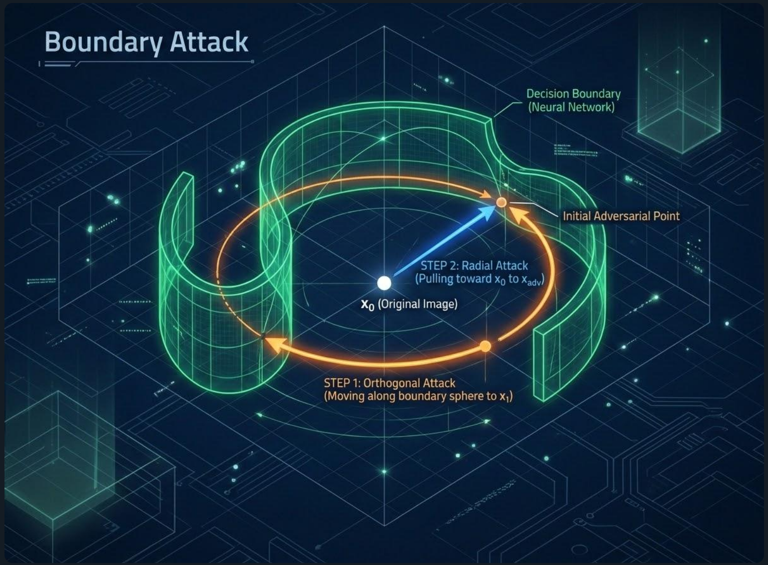
Fundamental Intuition: We find the closest point on the decision boundary to clean image x_0 by exploring the "wall" sideways.

TWO-STEP ITERATION:

- **1. Orthogonal Step (The Sidestep):** We step along a circle of constant distance. This explores the boundary without moving further away from x_0.
- **2. Radial Step (The Pull):** Once we find a better angle, we "pull" the sample closer to x_0.

The sidestep explores the shape of the wall, and the pull squeezes us closer whenever the wall allows it.

VISUALIZING THE ORTHOGONAL & RADIAL PROJECTION



PHASE 4: DIRECTION-BASED OPTIMIZATION

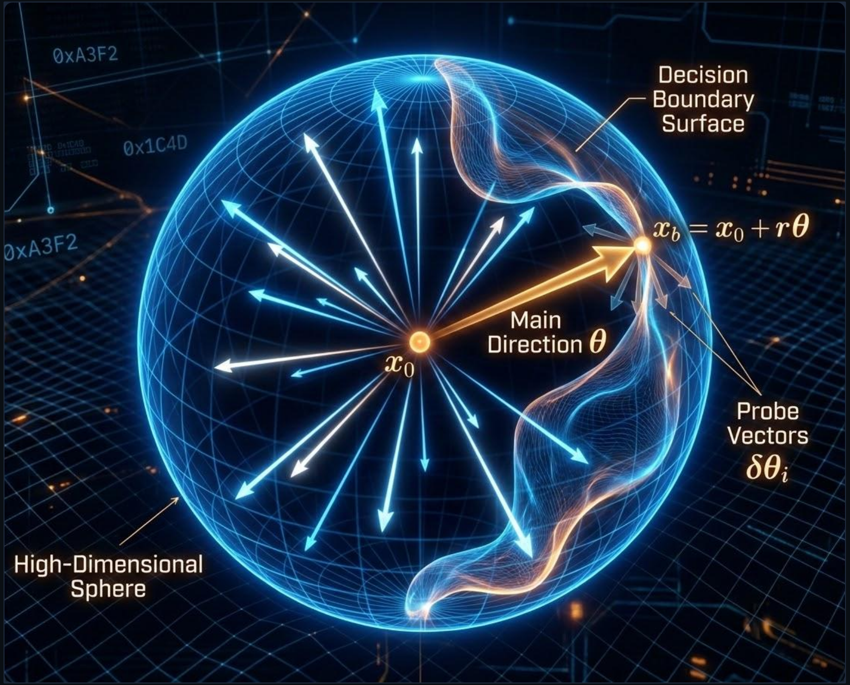
OPT Attack: The Angle-Based Distance Measurer

Concept: Unlike the Boundary Attack, the OPT family views the problem in terms of **angles (directions)** from the clean target x_0.

THE PROBING MECHANISM:

- **Directional Search:** For any direction θ, we calculate the boundary distance `g(θ)` using binary search.
- **Slope Estimation:** Nudges the compass with random directions `u` to see if the boundary gets closer.
- **The Update:** Averages probes to estimate the "steepest downhill slope" and turns θ accordingly.

The Goal: Find the unit direction vector θ that makes the boundary distance as small as possible.



Sign-OPT: The "Yes or No" Shortcut

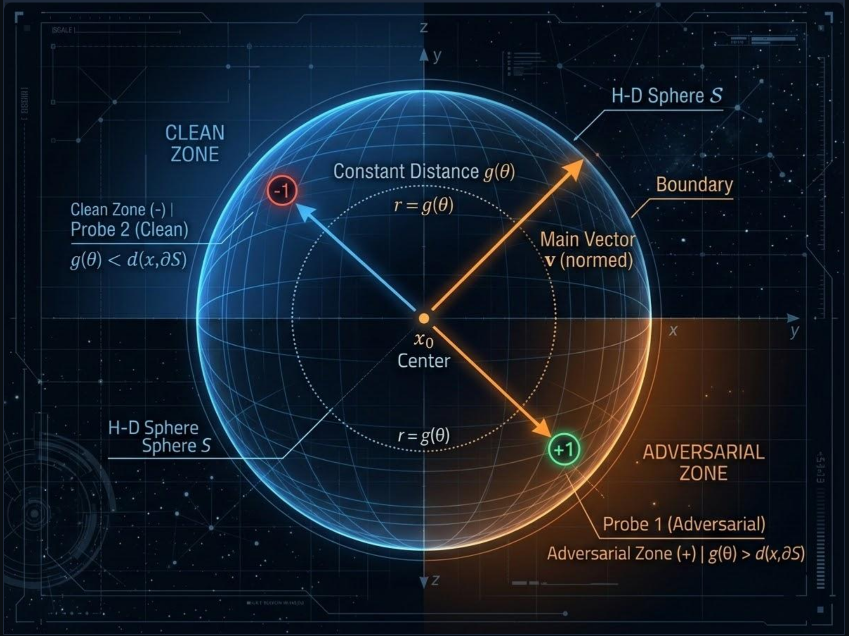
The Major Bottleneck: OPT runs a full binary search (7-10 queries) to test just one direction. Probing a step can cost **50+ queries**, wasting thousands over a full attack.

SIGN-BASED ESTIMATION:

- **The Test:** Instead of measuring exact distance, ask one question: "At the current distance $g(\theta)$, is this new direction still adversarial?"
- **Binary Result:** If **Yes** (Adversarial), the boundary moved closer (Sign=+1). If **No** (Clean), it moved farther (Sign=-1).
- s = 1.0 if is_adv(x0 + g_theta * u) else -1.0

Efficiency: Combining signs from q=12 probes estimates the gradient with **12 queries** vs OPT's 100+.

VISUALIZING THE SIGN-BASED GRADIENT PROBE



PHASE 5: BOUNDARY-BASED OPTIMIZATION

HopSkipJump: The Three-Step Dance

Concept: HopSkipJump (HSJA) estimates the gradient at the boundary by querying binary outcomes and projecting back to x_o.

THE ITERATIVE LOOP:

- **1. The Hop (Gradient Estimation):** Scatter 20 probes. Assign +1 if adversarial, -1 if clean. Average them to find the direction **g** perpendicular to the wall.
- **2. The Skip (The Step):** Take a step **x_step** into the adversarial region along vector **g**.
- **3. The Jump (Projection):** Binary search back towards target $\mathbf{x_o}$ to find a new boundary point closer than before.

Repeating this 15 times converges to the minimum distance adversarial sample.

VISUALIZING THE HOP-SKIP-JUMP GEOMETRY

Triangle Attack: Geometric Query Efficiency

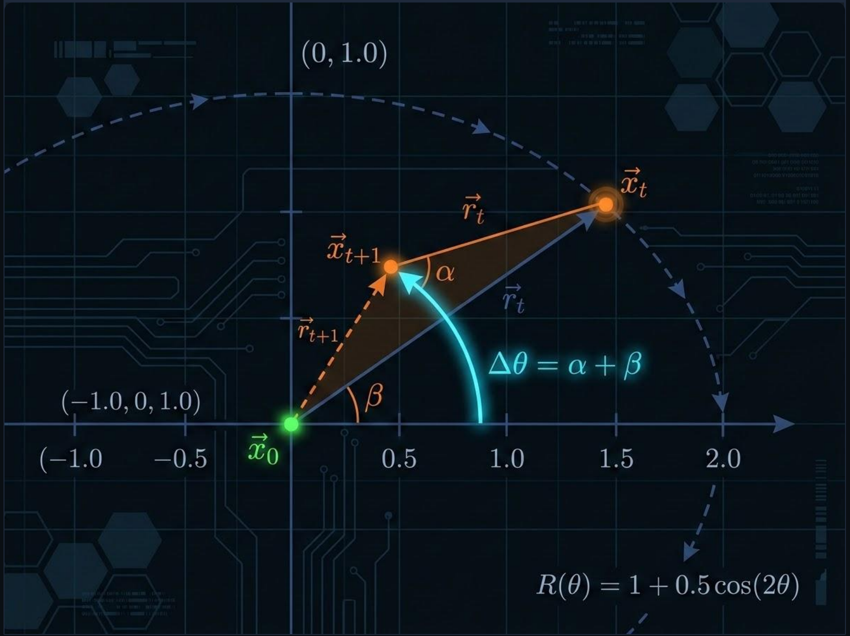
The Problem: The Query Bottleneck

HSJA and Sign-OPT spend 30-100 queries just to estimate one direction. The Triangle Attack replaces gradient estimation with the **Law of Sines**, using only 2-4 queries per step.

$$\delta_{t+1} = \delta_{t} \cdot [\sin(\alpha + \beta) / \sin(\alpha)]$$

- **The Geometry:** A 2D plane is defined via random DCT space, perpendicular to the target.
- **The Guarantee:** Math ensures $\delta_{t+1} < \delta_t$. We don't need gradients to know we are closer.
- **The Test:** Propose two candidates (rotated by ±β). If adversarial, move immediately.

VISUALIZING THE TRIANGLE ATTACK GEOMETRY



PHASE 5: BOUNDARY-BASED OPTIMIZATION

Biased Boundary Attack: The Hiker with a Guide Dog

The Efficiency Problem: Random Darts

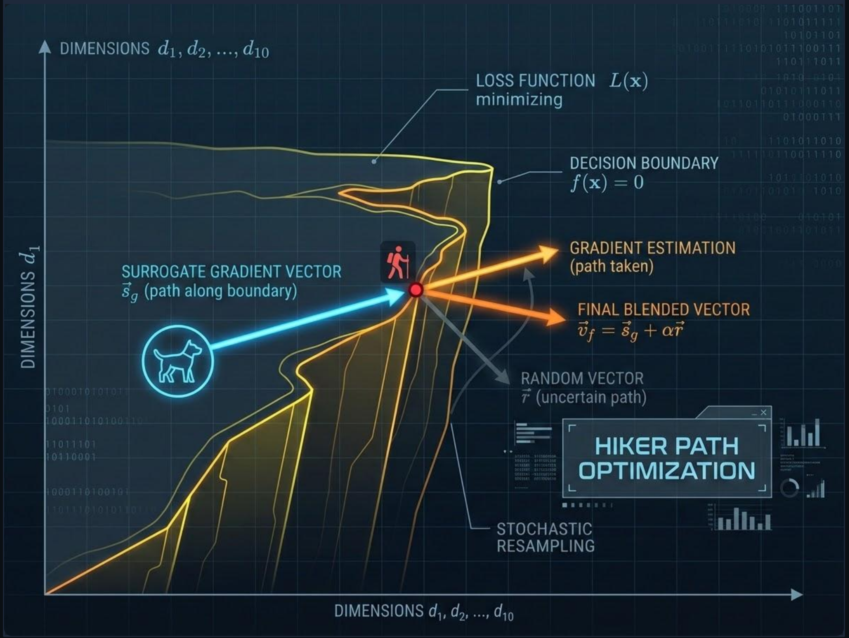
In 3072D space, random sidesteps are inefficient. You almost always "fall off the cliff," wasting queries. A biased approach provides a guide.

- **The Guide Dog (Surrogate):** A local model provides experience-based gradients to find the camp without falling.
- **The Blended Step:** Combines exploration with guidance.

```
eta = (1 - bias) * eta + bias * ctx.sgrad(x_b, y0)
```

By blending 50% random and 50% surrogate directions, proposed steps stay on path, reaching base camp in far fewer queries.

VISUALIZING THE BIASED GRADIENT BLEND



PHASE 5: BOUNDARY-BASED OPTIMIZATION

SQBA: The Best of Both Worlds

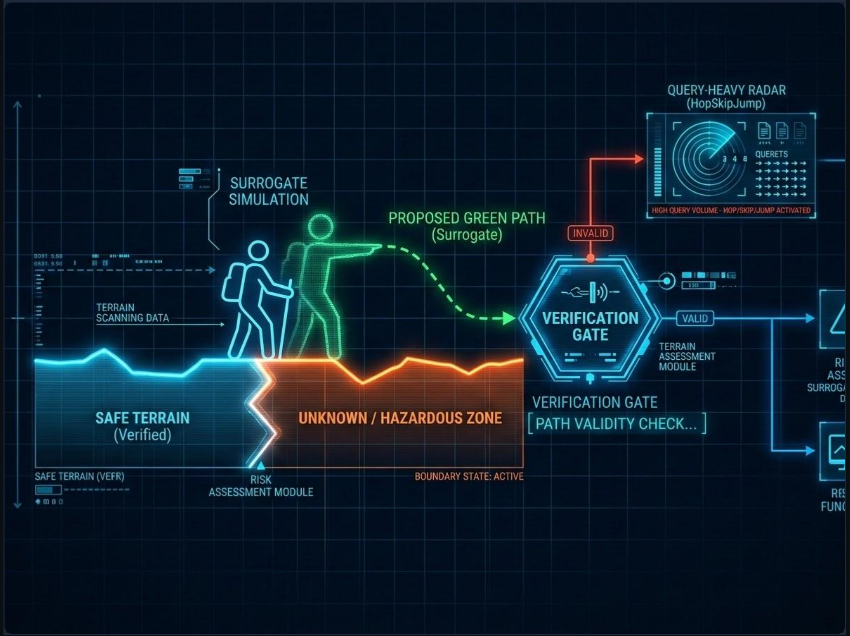
Concept: Trust but Verify

SQBA asks: "Why spend queries estimating a normal when a local surrogate can do it for free?"

- **1. The Free Trial:** Query the local surrogate gradient `ctx.sgrad(x_b,y0)` at 0 query cost.
- **2. The Verification:** Step along that direction and ask the victim: "Am I still adversarial?"
- **3. The Result:** If yes, we save 300+ queries. If no, we fall back to standard HopSkipJump estimation.

SQBA combines the **speed** of Transfer Attacks with the **guarantee** of Query-Based Attacks.

VISUALIZING THE SQBA SURROGATE HYBRID



Adversarial Attacks Have Shifted from White-Box to Black-Box Scenarios

The landscape of adversarial machine learning has systematically moved toward more realistic, restrictive threat models over the past decade.

Phase 1: 2014-2017

White-Box Dominance

Attackers required full access to model weights, architecture, and gradients. Methods like **FGSM (2014)** and PGD established the theoretical vulnerability of deep neural networks.

Phase 2: 2016-2018

Soft-Label & Transferability

Attackers utilized substitute models and prediction probabilities to craft **Transfer Attacks (2016)** without direct architectural access, proving that vulnerabilities transfer across models.

Phase 3: 2018-Present

Hard-Label Black-Box

Attackers only observe final class decisions. The evolution from **Boundary Attack (2018)** to **SQBA (2024)** represents continuous optimization of query efficiency under extreme constraints.

FGSM Proves Linear Behavior Causes Vulnerability

The Fast Gradient Sign Method (FGSM) revolutionized adversarial machine learning by proving neural networks are vulnerable due to their linear nature.

Core Mechanism: Computes the loss gradient with respect to the input and applies a small perturbation in the gradient sign direction.

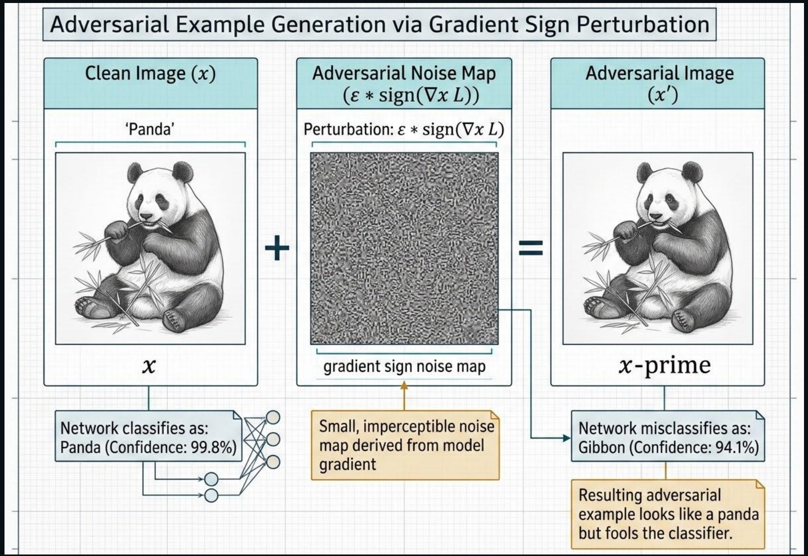
Mathematical Foundation:

$$x' = x + \varepsilon * \mathrm{sign}(\nabla_x L(\theta, x, y))$$

Efficiency: A single-step attack, computationally inexpensive and a foundational baseline.

Impact: Proved visually imperceptible changes reliably cause misclassification with high confidence.

FIGURE 1: GRADIENT SIGN PERTURBATION FLOW



PGD: The Ultimate First-Order Adversary

Projected Gradient Descent (PGD) was introduced as a **universal first-order adversarial attack**, significantly strengthening the evaluation of model robustness.

FGSM (FAST GRADIENT SIGN METHOD)

SINGLE-STEP NATURE

FGSM takes a **single step** in the direction of the gradient sign. It is computationally efficient but often suboptimal, as the step can easily overshoot the decision boundary or exit the valid epsilon-ball constraint.

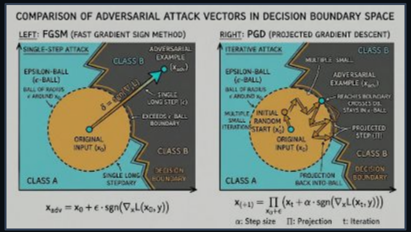
```
x_adv = x_0 + ε · sgn(∇_x L(x_0, y))
```

VS

PGD (PROJECTED GRADIENT DESCENT)

ITERATIVE & ROBUST

- **Iterative Optimization:** Applies multiple smaller steps, projecting back into the ε-ball after each iteration.
- **Random Initialization:** Restarts at random points within the ε-ball to avoid local maxima.
- **Guarantee:** Defending against PGD provides empirical robustness against al



GEOMETRIC PATH COMPARISON

Transferability Enables Practical Black-Box Exploits

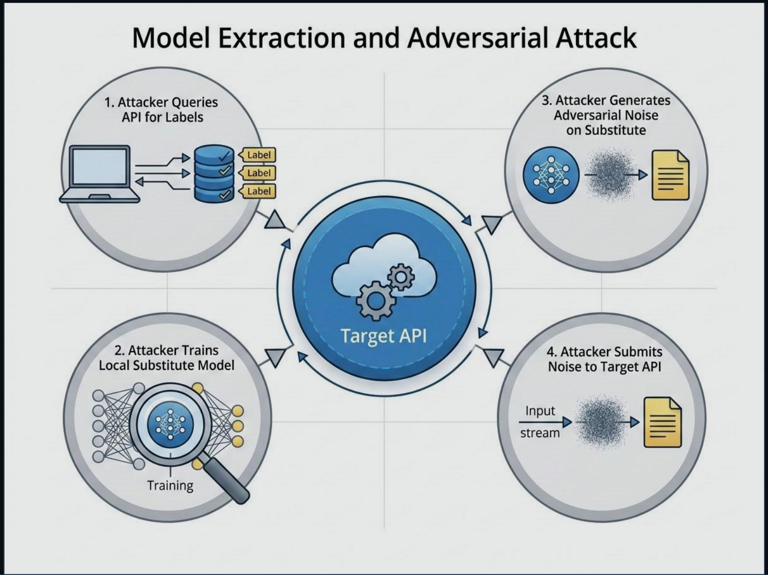
Substitute Model Strategy

Attackers query a target model (e.g., cloud API) to label a synthetic dataset, then train a local **'substitute' model**.

Cross-Model Transfer

- **White-Box Exploitation:** Attackers use gradient methods (FGSM/PGD) on their own substitute model.
- **Reliable Transfer:** Examples generated on the substitute reliably fool targets with different architectures (e.g., DT to DNN).
- **Real-World Impact:** Successfully bypassed commercial APIs from Amazon and Google.

FIGURE 1: HUB-AND-SPOKE EXPLOIT WORKFLOW

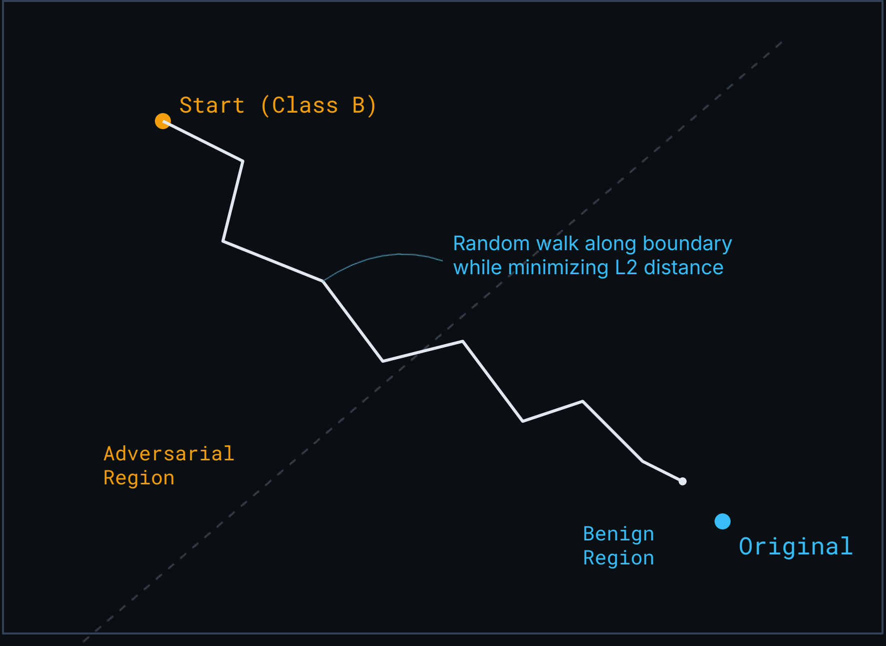


Hard-Label Black-Box Attacks Are Feasible

The **Boundary Attack** (Brendel et al.) pioneered decision-based models, requiring only the final class label rather than probabilities.

- **Initialization:** Starts with an image from the target class that is fundamentally adversarial to the model.
- **Random Walk Optimization:** Iteratively performs a random walk along the decision boundary.
- **Distance Minimization:** Reduces L2 distance to the original image while staying on the adversarial side.
- **Query Intensity:** Conceptually simple but requires millions of API calls to converge.

FIGURE 1: BOUNDARY ATTACK TRAJECTORY



OPT and Sign-OPT: Continuous Optimization

The OPT and Sign-OPT frameworks dramatically improved query efficiency by transforming the discrete hard-label problem into a **continuous optimization task**.

CONTINUOUS FORMULATION & ZEROTH-ORDER

- **Distance Optimization:** Formulates the attack by searching for the minimum distance to the decision boundary along specific search directions.
- **Zeroth-Order Estimation:** Evaluates the function at multiple points to estimate the gradient of the distance function when true gradients are unavailable.

VS

SIGN-OPT REFINEMENT & EFFICIENCY

- **Directional Derivative:** Improved efficiency by only estimating the **sign** of the directional derivative rather than precision magnitude.
- **Efficiency Gains:** Reduced query count by up to **80%** compared to the original Boundary Attack.
- **Practicality:** Makes hard-label attacks viable against real-world rate-limited systems.

Performance Metric

QUERY_REDUCTION ≈ 80% (vs. Boundary Attack)

HopSkipJump: Refined Gradient Estimation

HopSkipJumpAttack is a hyperparameter-free approach to decision-based adversarial examples.

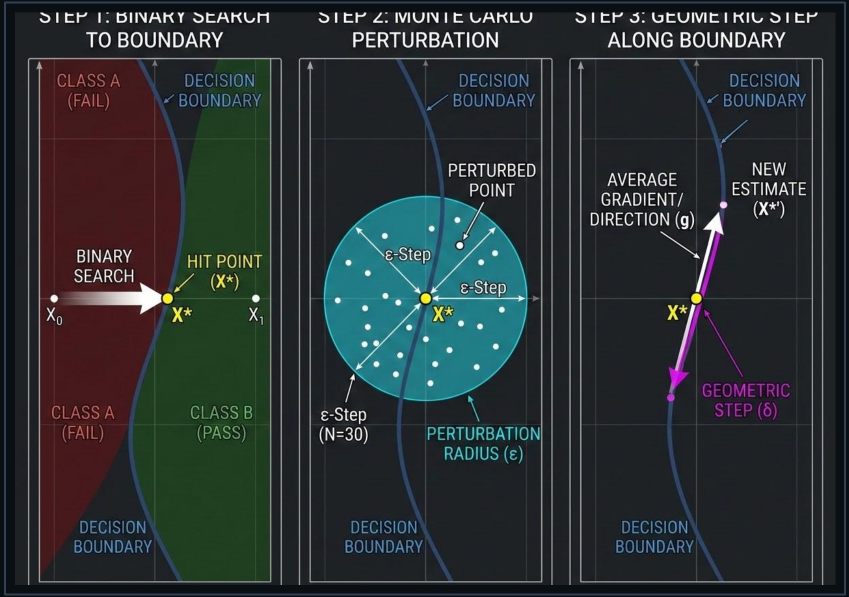
Binary Search Initialization: Precisely locates the decision boundary between classes.

Gradient Estimation: Queries the model with Monte Carlo sampled perturbations.

Geometric Progression: Moves along the boundary to minimize adversarial distance.

Hyperparameter-Free: Robust plug-and-play solution that automatically adapts step sizes.

FIGURE 1: BOUNDARY ESTIMATION WORKFLOW

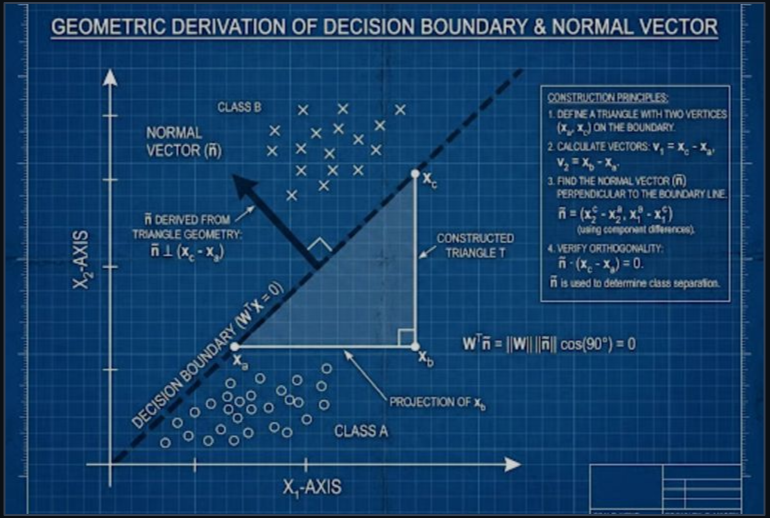


Triangle Attack: Geometric Query Efficiency

The Triangle Attack introduces a geometric perspective to hard-label black-box attacks, drastically reducing the query budget for imperceptibility.

- **Geometric Gradient Estimation:** Uses triangle properties from three points near the boundary to infer the normal vector (gradient).
- **Dimensionality Reduction:** Targets low-frequency components in a reduced-dimension subspace, maximizing vulnerability.
- **Query Efficiency:** Requires significantly fewer evaluations than HopSkipJump to achieve the same L2 distance.
- **Practical Application:** Viable against commercial APIs with strict rate limits and cost-per-query models.

FIGURE 1: TRIANGLE GEOMETRIC INFERENCE



Biased Boundary: Guessing Smart

The Biased Boundary Attack combines white-box transferability with decision-based precision to **accelerate convergence**.

HYBRID APPROACH & BIASING

Surrogate Gradients: Uses imperfect gradients as directional guides for black-box queries.
Biased Distribution: Perturbations are heavily weighted toward the white-box gradient vector.
Standard vs Biased: Replaces uniform random walk with targeted search steps.

Uniform Random Walk (Original)

VS

ACCELERATED CONVERGENCE

Low-Frequency Focus: Biases toward spatial domains likely to cross architectures.
Dead-End Reduction: Drastically reduces the number of queries that fail to stay adversarial.
Efficiency: Rapidly travels along the boundary towards the target class center.

Surrogate-Biased Walk (Brunner et al.)

SMALL QUERY BLACK-BOX ADVERSARIAL ATTACK (2024)

SQBA: The Pinnacle of Query Efficiency

SQBA represents the current state-of-the-art in query-efficient hard-label attacks, synthesizing a decade of advancements into a highly optimized framework.

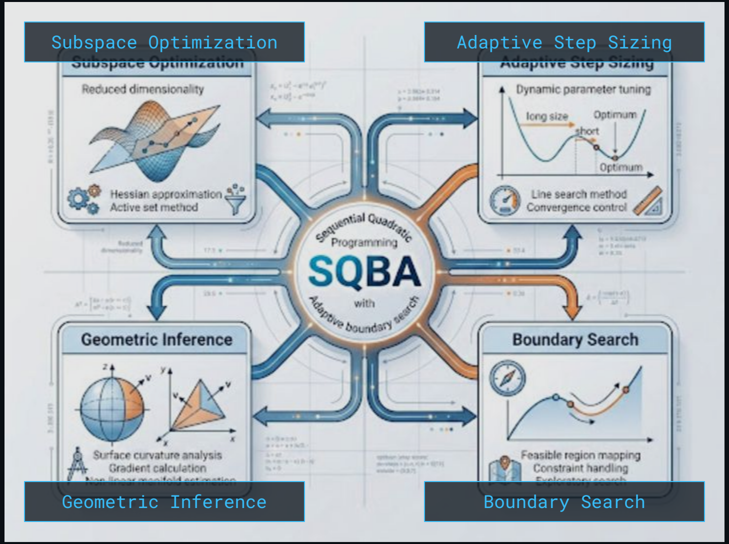
Extreme Query Efficiency: Designed for severe query budgets to evade detection mechanisms and rate limits.

Subspace Optimization: Leverages dimensionality reduction, optimizing perturbations in transferable subspaces.

Adaptive Step Sizing: Search parameters adjust dynamically based on decision boundary curvature.

Culmination of Lineage: Combines HopSkipJump, Triangle Attack, and surrogate biasing into one protocol.

FIGURE 1: SYNTHESIZED ARCHITECTURE

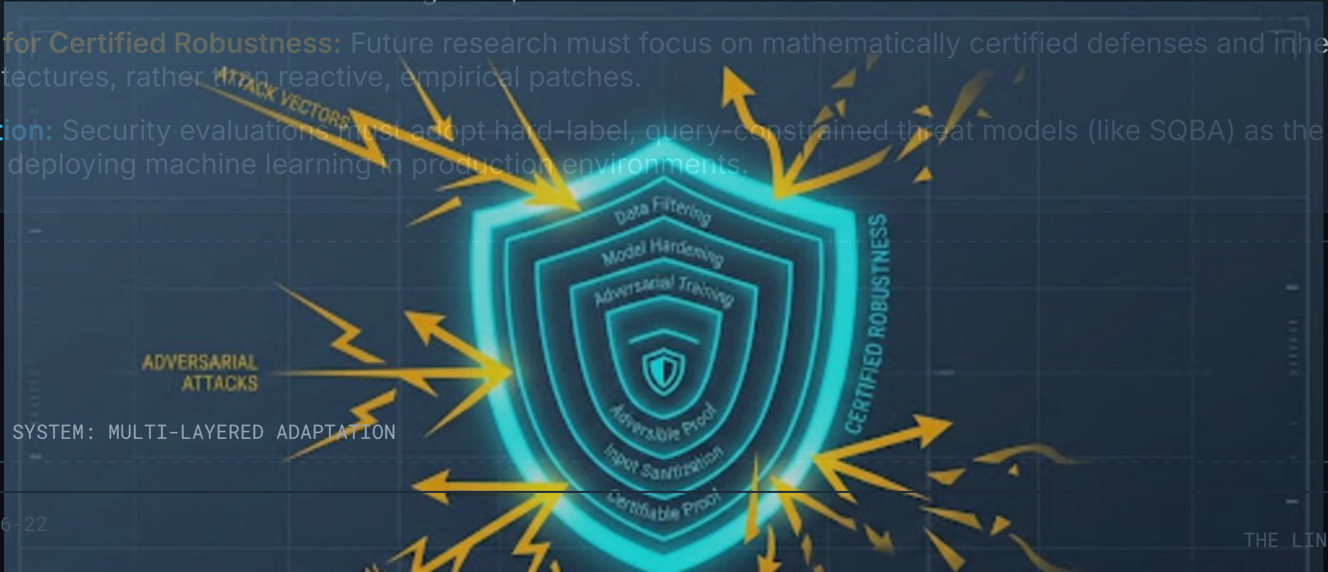


CONCLUSION: THE EVOLUTION AND FUTURE DIRECTIONS

Future Robustness: Beyond Gradient Masking

The decade-long evolution from FGSM to SQBA demonstrates that restricting attacker access does not guarantee model security.

- **The Failure of Obfuscation:** The transition to hard-label attacks proves that hiding probabilities or masking gradients is an insufficient defense. Attackers can always optimize around the decision boundary.
- **The Query Efficiency Arms Race:** As attacks like SQBA require fewer queries, rate-limiting and standard API protections become less effective. Attacks are becoming cheaper and stealthier.
- **The Need for Certified Robustness:** Future research must focus on mathematically certified defenses and inherently robust architectures, rather than reactive, empirical patches.
- **Call to Action:** Security evaluations must adopt hard-label, query-constrained threat models (like SQBA) as the standard baseline for deploying machine learning in production environments.



ROBUSTNESS DEFENSE SYSTEM: MULTI-LAYERED ADAPTATION